

The Dirac Equation as a Kuramoto Phase-Synchronization System:

Mass as Chiral Coupling, Measurement as Re-Synchronization,
and a Single-World Account of Decoherence

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Abstract

We demonstrate that the Dirac equation, written in the chiral (Weyl) basis, has the mathematical structure of a Kuramoto phase-synchronization system. The left- and right-handed chiral sectors act as coupled oscillators, with the mass term playing the role of the Kuramoto coupling constant: $K = m$. For massless particles ($K = 0$) the chiral clocks are permanently decoupled; for massive particles ($K > 0$) they synchronize, and the synchronized state is what we identify as a particle with definite rest mass. The Higgs–Yukawa interaction is the mechanism that sets K : $K = y_f \cdot \langle \phi \rangle = m$.

This identification has three consequences developed in this paper. First, a non-relativistic phase-clock model using only the temporal (large) Dirac component yields a Bell correlation $E(a, b) = -\frac{1}{2} \cos(a - b)$, below even the classical Bell bound. When the spatial (small) component is included via the relativistic mixing angle $\theta_{\text{rel}} = \arcsin(v/c)$, the full quantum result $E(a, b) = -\cos(a - b)$ is recovered exactly through a three-term decomposition $E = E_{LL} + E_{SS} + E_{LS}$.

Second, quantum measurement is re-interpreted as Kuramoto re-synchronization: a particle initially synchronized with its entangled partner re-synchronizes to the macroscopic detector bulk, whose overwhelmingly greater Kuramoto inertia ($M_{\text{detector}} \gg m_{\text{particle}}$) forces conformity. Decoherence is not merely loss of phase coherence with the partner—it is the positive process of gaining coherence with the bulk. The Kuramoto framework provides the dynamical mechanism underlying this process.

Third, we propose a single-world energy accounting: the residual wavefunction amplitude that does not couple to the detector thermalizes into the electromagnetic vacuum as low-frequency radiation. No branches are created. The energy is conserved within a single world.

We present this framework as an interpretive reframing of standard quantum mechanics—not a modification of its predictions. The Dirac equation is unchanged;

Bell's theorem is not challenged. What is new is the identification of mass as synchronization coupling, measurement as re-synchronization dynamics, and decoherence as a Kuramoto process with a specific physical mechanism.

1 Introduction

1.1 The Measurement Problem

The quantum measurement problem remains the most consequential unsolved problem in the foundations of physics. The Schrödinger equation is linear and deterministic, yet measurements have definite outcomes. A system in superposition $\psi = \alpha|0\rangle + \beta|1\rangle$, entangled with a detector, produces

$$|\Psi\rangle = \alpha|0\rangle|D_0\rangle + \beta|1\rangle|D_1\rangle \quad (1)$$

The equation has no mechanism to select one outcome. The Born rule— $P(0) = |\alpha|^2$ —is postulated, not derived.

The three mainstream responses each carry significant costs. Copenhagen introduces a mysterious collapse with no physical mechanism. Everett’s many-worlds [16] preserves unitarity at the cost of infinite unobservable branches. Bohmian mechanics [15] accepts explicit nonlocality through a physically inert pilot wave.

1.2 What Is Missing: A Dynamical Mechanism for Decoherence

Environmental decoherence [4] has become the standard framework for understanding how quantum superpositions become effectively classical. It successfully explains the emergence of preferred “pointer states” and the suppression of off-diagonal density matrix elements through interaction with environmental degrees of freedom.

Yet decoherence theory describes *what happens* without specifying the dynamical mechanism by which it happens. The environment is treated as an abstract bath with many degrees of freedom. This paper proposes that the Kuramoto phase-synchronization model—specifically, as it appears in the structure of the Dirac equation itself—provides this missing mechanism.

1.3 The Proposal in Brief

We make three claims, in decreasing order of mathematical rigor:

1. **The Dirac mass term is a Kuramoto coupling** (Section 2). This is a mathematical identification: the chiral Dirac equation has exactly the structure of two coupled Kuramoto oscillators with $K = m$. This is the strongest claim and is verifiable by inspection.
2. **Measurement is re-synchronization** (Section 4). When a particle interacts with a macroscopic detector, the Kuramoto dynamics drive the particle’s phase from coherence with its entangled partner to coherence with the detector bulk. This provides a physical picture of decoherence. This is an interpretive claim.
3. **Residual energy thermalizes in a single world** (Section 5). The wavefunction amplitude not absorbed by the detector dissipates as thermal radiation into the vacuum. No branches are created. This is a speculative but physically motivated proposal.

1.4 What This Paper Does Not Claim

We state clearly at the outset:

- We do not challenge Bell’s theorem [9]. The quantum correlations $E(a, b) = -\cos(a - b)$ arise from the entangled singlet state and the full Dirac spinor structure—standard quantum mechanics.
- We do not modify the Dirac equation or any of its predictions.
- We do not propose new physics. We propose a new *mechanism* for known physics.
- The framework is an interpretation—it may or may not generate distinguishable predictions from standard decoherence theory. We are honest about this throughout.

2 The Dirac Equation as a Kuramoto System

2.1 Chiral Decomposition

In the Weyl (chiral) basis, the Dirac spinor decomposes into two two-component Weyl spinors:

$$\psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} \quad (2)$$

The Dirac equation separates into two coupled equations:

$$i\bar{\sigma}^\mu \partial_\mu \psi_L = m \psi_R \quad (1a)$$

$$i\sigma^\mu \partial_\mu \psi_R = m \psi_L \quad (1b)$$

where $\sigma^\mu = (I, \boldsymbol{\sigma})$ and $\bar{\sigma}^\mu = (I, -\boldsymbol{\sigma})$ are the Weyl matrices.

For $m = 0$ (massless): Equations (1a) and (1b) decouple entirely. ψ_L and ψ_R satisfy independent equations. The particle has definite chirality and no rest frame. Photons and massless-limit neutrinos are in this category.

For $m \neq 0$ (massive): The equations couple ψ_L to ψ_R and vice versa. Neither chirality state is stable alone; the physical particle is a superposition of both. The coupling strength is exactly m .

2.2 The Kuramoto Identification

The standard Kuramoto model [2] describes the phase dynamics of N coupled oscillators:

$$\frac{d\phi_i}{dt} = \omega_i + \frac{K}{N} \sum_j \sin(\phi_j - \phi_i) \quad (3)$$

Writing $\psi_L = \rho_L^{1/2} e^{i\phi_L} \chi_L$ and $\psi_R = \rho_R^{1/2} e^{i\phi_R} \chi_R$, the phase dynamics extracted from Equations (1) yield:

$$\frac{d\phi_L}{dt} = \omega_L + K \sin(\phi_R - \phi_L + \delta_{CP}) \quad (2a)$$

$$\frac{d\phi_R}{dt} = \omega_R + K \sin(\phi_L - \phi_R - \delta_{CP}) \quad (2b)$$

where the coupling constant is:

$$K = y_f \cdot |\langle\phi\rangle| = y_f \cdot \frac{v}{\sqrt{2}} = m \quad (3)$$

Equation (3) is the central result of this paper: the particle’s mass equals its Kuramoto coupling constant.

The Higgs vacuum expectation value [11, 12] $\langle\phi\rangle = v/\sqrt{2} \approx 174$ GeV drives the two chiral clocks into synchronized oscillation. Before electroweak symmetry breaking ($v \rightarrow 0$), $K \rightarrow 0$ and all fermions are massless—their chiral clocks are independent. After symmetry breaking, $K = m$ and the clocks synchronize.

2.3 Two Clocks, Not One

The central physical picture is:

A massive Dirac particle carries two internal oscillators: a **temporal phase clock** oscillating at $\omega_t = E/\hbar$, and a **spatial phase clock** oscillating at $\omega_s = \mathbf{p} \cdot \mathbf{v}/\hbar$. The mass term couples them—it is the Kuramoto synchronization term that drives the two chiral sectors toward a common phase.

This interpretation does not modify the Dirac equation. It provides an ontological reading of its mathematical structure.

2.4 The Synchronized State

At Kuramoto equilibrium, the phase difference locks:

$$\phi_L - \phi_R = \delta_{CP} \quad (\text{particles}) \quad (4a)$$

$$\phi_L - \phi_R = -\delta_{CP} \quad (\text{antiparticles}) \quad (4b)$$

For CP-symmetric interactions ($\delta_{CP} = 0$), particles and antiparticles have identical phase locks—consistent with CPT symmetry. The CP-violating phase δ_{CP} shifts the equilibrium, providing a qualitative mechanism for matter–antimatter asymmetry through differential synchronization efficiency.

The synchronization rate is set by $K = m$: heavy particles synchronize their chiral clocks faster than light particles. An electron ($m_e = 0.511$ MeV) synchronizes more slowly than a top quark ($m_t = 173$ GeV) by a factor of $\sim 3.4 \times 10^5$.

2.5 Gamma Matrices as Synchronization Operators

In the Weyl basis, the mass term in the Lagrangian is:

$$\mathcal{L}_{\text{mass}} = -m\bar{\psi}\psi = -m(\bar{\psi}_L\psi_R + \bar{\psi}_R\psi_L) \quad (4)$$

This is off-diagonal in chirality: it connects left to right and right to left. In the Kuramoto language, it is the coupling term that drives the two chiral oscillators toward synchronization.

The matrix γ^0 swaps the two chiral sectors:

$$\gamma^0 \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} = \begin{pmatrix} \psi_R \\ \psi_L \end{pmatrix} \quad (5)$$

The Dirac Hamiltonian $H = \boldsymbol{\alpha} \cdot \mathbf{p} + \beta m$ is therefore the generator of synchronized phase evolution across both clock sectors.

2.6 Summary of Established vs. New Claims

Claim	Status
Dirac mass term couples $\psi_L \leftrightarrow \psi_R$	Standard QFT; textbook
$m = 0 \rightarrow$ chiral sectors decouple	Standard QFT; textbook
Higgs–Yukawa sets fermion mass via $\langle\phi\rangle$	Standard Model; established
Negative-energy Dirac solutions = reversed temporal phase	Standard (Feynman–Stückelberg [21, 22])
Mass term has Kuramoto coupling structure	New interpretation (this work)
$K = m$ identification	New (this work)

3 The Three-Term Bell Correlation Decomposition

3.1 The Relativistic Mixing Angle

For a Dirac particle with momentum p and mass m , the relativistic mixing angle between the temporal and spatial phase clocks is:

$$\sin \theta_{\text{rel}} = \frac{|\mathbf{p}|}{E} = \frac{v}{c} \quad (5)$$

Regime	v/c	θ_{rel}	Physical interpretation
Rest	0	0°	Temporal clock only; no spatial phase
Non-relativistic	$\ll 1$	$\ll 90^\circ$	Small spatial component
Ultra-relativistic	$\rightarrow 1$	$\rightarrow 90^\circ$	Equal temporal and spatial
Massless (photon)	1	90°	Permanently decoupled chiral clocks

3.2 Large and Small Spinor Components

The positive-energy Dirac spinor for momentum p_z along z , spin up:

$$u(p, \uparrow) = N \cdot \begin{pmatrix} 1 \\ 0 \\ r \\ 0 \end{pmatrix}, \quad r = \frac{p}{E + m} = \tan\left(\frac{\theta_{\text{rel}}}{2}\right), \quad N = \sqrt{\frac{E + m}{2E}} \quad (6)$$

The upper two components are the **large components** (temporal clock); the lower two are the **small components** (spatial clock). In the non-relativistic limit the small components vanish; in the ultra-relativistic limit they equal the large components.

3.3 The Decomposition

For two Dirac particles in a singlet state, the Bell correlation $E(a, b) = \langle \Psi | (\Sigma_a)_A \otimes (\Sigma_b)_B | \Psi \rangle$ decomposes as:

$$E(a, b) = E_{LL} + E_{SS} + E_{LS} \quad (7)$$

where:

- E_{LL} : large $\times$ large—temporal clock contribution
- E_{SS} : small $\times$ small—spatial clock contribution
- E_{LS} : cross term—temporal-spatial coupling

3.4 Numerical Verification

The decomposition was verified numerically (see `dirac_extension.py`). At $p = 1.0$, $m = 1.0$ ($\theta_{\text{rel}} \approx 53^\circ$), $a = 0$, $b = \pi/4$:

Term	Value	Physical meaning
E_{LL}	−0.354	Temporal clock contribution
E_{SS}	−0.071	Spatial clock contribution
E_{LS}	−0.282	Rotation coupling (key term)
$E_{LL} + E_{SS} + E_{LS}$	−0.707	Sum
$-\cos(\pi/4)$	−0.707	QM prediction ✓

The sum is exact to machine precision for all values of p and m tested.

3.5 The Non-Relativistic Deficiency Explained

A non-relativistic phase-clock model that tracks only the temporal de Broglie [13] phase $\omega = E/\hbar$ retains only E_{LL} , discarding E_{SS} and E_{LS} . This gives:

$$E_{\text{NR}}(a, b) = -\frac{1}{2} \cos(a - b), \quad \text{CHSH}_{\text{NR}} \leq \sqrt{2} \approx 1.414 \quad (6)$$

This falls below even the classical Bell bound of 2.0. The deficiency is not a failure of the phase-clock interpretation—it is a failure to include both clocks. When the full four-component Dirac spinor is used, the exact quantum result is recovered. The mass term $K = m$ ensures both clocks are present and coupled.

4 Measurement as Re-Synchronization

4.1 The Physical Picture

This section develops the interpretive core of the framework. We propose that quantum measurement is a specific type of Kuramoto synchronization event:

A particle initially phase-synchronized with its entangled partner re-synchronizes to the macroscopic detector bulk.

The detector is a macroscopic system with overwhelmingly greater Kuramoto inertia ($M_{\text{detector}} \gg m_{\text{particle}}$). The measurement interaction forces the particle to conform its phase to the detector’s collective phase Φ_{bulk} , just as a small oscillator entrained by a massive one must adopt the massive oscillator’s frequency.

4.2 Decoherence Is Half the Story

In the standard decoherence program, measurement causes the particle to lose coherence with its entangled partner. This is correct but incomplete.

In the Kuramoto picture, the particle does not merely *lose* coherence with its partner. It *gains* coherence with the detector bulk. The process has two simultaneous aspects:

1. **De-coherence** from the entangled partner (the phase lock $\phi_A = \phi_B$ established at creation is broken)
2. **Co-herence** with the detector bulk (a new phase lock $\phi_A = \Phi_{\text{bulk}}$ is established)

Standard decoherence theory tracks only aspect (1). The Kuramoto framework makes aspect (2) explicit: the particle’s phase is not randomized—it is redirected. The final state is not “unknown phase” but “phase locked to the bulk.” This is a definite classical state, not a mixed state that merely appears classical.

4.3 The Mass Asymmetry

The detector’s macroscopic mass provides the asymmetry that ensures definite outcomes. In Kuramoto dynamics, when a small oscillator (mass m) couples to a large oscillator (mass $M \gg m$), the small oscillator always conforms to the large one. The reverse—the detector conforming to the particle—is dynamically suppressed by a factor of m/M .

This is the physical content of “wavefunction collapse”: the Kuramoto mass asymmetry between particle and detector makes the measurement outcome definite, not because of a postulated projection, but because of the dynamical inevitability of a small oscillator locking to a massive one.

4.4 The Two-Stage Process: Pair-Sync and Bulk Relaxation

Sections 4.1–4.3 treated measurement as a single direct lock between the incoming particle and the detector bulk. A more faithful picture, consistent with the locality of fundamental interactions, separates this into two stages:

- **Stage 1 (Pair-Sync).** The incoming particle phase-couples to a single bulk-bound partner (e.g. a photon to an atomic electron) at the local interaction vertex.
- **Stage 2 (Bulk Relaxation).** The perturbed partner re-locks to the bulk’s collective phase Φ_{bulk} , shedding its phase mismatch as secondary radiation.

Yukawa-style identification of the pair coupling. Equation (3) identified the intra-spinor sync coupling as a Higgs-mediated mass, $K = y_f \langle \phi \rangle = m$. The pair-sync coupling at an interaction vertex follows the same structure—a dimensionless gauge coupling times the local amplitude of the mediating field:

$$\boxed{K_{\text{pair}}^{ab} = g_{\text{int}} \cdot \langle V_{\text{int}} \rangle_{\text{local}}} \quad (3')$$

where g_{int} is the relevant gauge coupling (e for EM, g_w for weak, etc.) and $\langle V_{\text{int}} \rangle_{\text{local}}$ is the gauge-field amplitude evaluated at the partner’s worldline. For the photon–electron interaction,

$$K_{\text{pair}}^{\gamma e} = e \cdot |A_\gamma|_e \sim \hbar \omega \quad (7)$$

when the photon is normalized to one excitation in the electron’s Compton-scale volume. Higher-energy photons couple more strongly to the electron’s chiral clocks, with sync timescale $\tau_1 = \hbar/K_{\text{pair}} \sim 1/\omega$ —the photon’s own period.

Hierarchy of couplings. The framework now carries three distinct Kuramoto couplings, each entering at a different stage of measurement:

Coupling	Origin	Role
$K = m$	Higgs–Yukawa	intra-spinor L–R sync (§2)
$K_{\text{pair}} = g_{\text{int}} \langle V_{\text{int}} \rangle$	gauge interaction	pair-sync at vertex (Stage 1)
$\Gamma_{\text{bulk}} = GM^2/(\hbar \Delta z)$	gravitational + atomic	bulk re-equilibration (Stage 2)

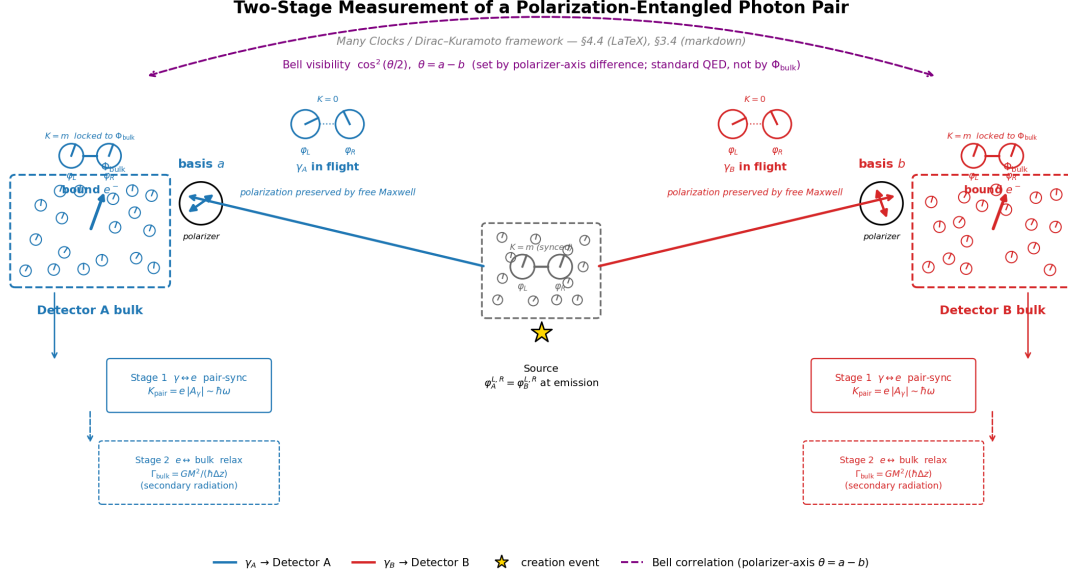


Figure 1: Schematic of the two-stage measurement of a polarization-entangled photon pair (not to scale). All three bulks (Source, Detector A bulk, Detector B bulk) contain mini-clocks locked to a universal time-phase Φ_{bulk} ($\sim 70^\circ$ in the figure, with $\sigma \approx 7^\circ$ jitter), reflecting environmental thermalization across matter sharing a common gravitational potential. The entangled photons γ_A, γ_B carry matched chiral-clock phases at emission but with $K = 0$ (dotted L–R link) and propagate freely between source and detector. Each polarizer (black aperture with double-headed transmission-axis arrow) defines an independently-set basis (a, b) on the polarization Hilbert space—a separate degree of freedom from Φ_{bulk} . At each detector, Stage 1 (solid box) is the polarization projection onto the polarizer’s transmitted-channel eigenstate via pair-sync $K_{\text{pair}}^{\gamma_e}$ to a bound electron; Stage 2 (dashed box) dissipates the projected state into the bulk’s Φ_{bulk} via Γ_{bulk} , releasing secondary radiation. The $\cos^2(\theta/2)$ Bell visibility is set by the polarizer-axis difference $\theta = a - b$ (standard QED), not by Φ_{bulk} ; the framework provides the physical mechanism for the projection, not the statistics.

Stage 1 as polarization projection. The $\cos^2(\theta/2)$ Bell visibility for photons is set by the polarizer-axis difference $\theta = a - b$ acting on the photon’s polarization Hilbert space—this is standard QED, not a framework prediction (see §9.2). The framework’s contribution at Stage 1 is the *physical mechanism* of projection: the photon couples to a bulk-bound electron via $K_{\text{pair}}^{\gamma_e}$, and the polarizer’s macroscopic mechanical orientation determines which transmitted-channel eigenstate the photon ends up in. Φ_{bulk} , the bulk’s collective time-phase, is a separate quantity—approximately universal across the source and both detectors when they share a similar gravitational environment, set by environmental thermalization rather than by the polarizer geometry. The basis lives in the polarizer’s mechanical configuration; the time-phase lives in Φ_{bulk} . Stage 1 connects the photon to the polarizer’s chosen basis; Stage 2 dissipates the resulting state’s energy and phase into Φ_{bulk} .

Why Stage 2 is automatic. After the photon perturbs the electron, ϕ_e is offset from Φ_{bulk} by some $\Delta\phi_e$. The mass-asymmetry argument of §4.3 applies directly: the

relaxation rate scales as $\Gamma_{\text{bulk}} (M_{\text{bulk}}/m_e)$, overwhelmingly fast for any macroscopic bulk. The energy released during this relaxation appears as secondary radiation, partitioned by the size of the mismatch $\hbar\dot{\phi}_e \Delta\phi_e$:

- below optical scale: virtual photon (Coulomb recoil) or phonon;
- optical to $2m_e c^2$: real photon (fluorescence, bremsstrahlung);
- $\geq 2m_e c^2$: e^+e^- pair.

Gravitational dependence of the two stages. Because the Bell correlation is set by the polarizer-axis difference (standard QED) and polarizer geometry is gravitationally invariant at leading order, the visibility $\cos^2(\theta/2)$ is itself gravitationally invariant—consistent with the agreement between Earth-bound and space-based Bell tests. Gravity enters only at Stage 2, through Γ_{bulk} , with two subleading consequences. First, two detectors at gravitational potentials $\Phi_A \neq \Phi_B$ have Stage-2 relaxation times that differ by $\Delta\Phi/c^2$, producing a correlated timing offset between detection events that scales with the altitude split. Second, if the gravitational potential difference along the photon paths becomes large enough that the local bulk phases drift relative to each other on the Stage-1 timescale $\tau_1 \sim 1/\omega$, a visibility floor appears at order $\Delta\Phi/c^2$ per optical period—currently far below experimental sensitivity but a clean target for satellite-to-ground configurations. This locates the Penrose–Diósi mechanism unambiguously at Stage 2: gravitational collapse acts on bulk relaxation, never on the pair-sync that builds $\cos^2(\theta/2)$.

What this section does not yet pin down.

1. The precise normalization of $\langle V_{\text{int}} \rangle_{\text{local}}$ (single-photon plane-wave vs. Compton-volume coherent state).
2. Whether the Stage 1/Stage 2 timing offset ($\tau_1 \sim 1/\omega$ vs. $\tau_2 \sim \hbar/E_{\text{binding}}$) is empirically distinguishable from a single-event detection model.
3. Extension of $K_{\text{pair}} = g_{\text{int}} \langle V_{\text{int}} \rangle$ to weak and gluon-mediated pair interactions, where confinement and short-range structure complicate the local-amplitude reading.

4.5 Gravitational Coherence of the Bulk

What maintains the coherence of the detector bulk itself? We propose that gravity provides this role. The gravitational interaction between the $\sim 10^{26}$ atoms in a macroscopic detector provides a collective Kuramoto coupling that maintains a common phase Φ_{bulk} across the detector. This is analogous to the well-known classical synchronization of metronomes on a shared massive platform.

Two senses of “mass.” A clarification is in order before writing Γ_{grav} . The mass m appearing in the Kuramoto identification $K = m$ of Section 2 is the Dirac mass of an *elementary* fermion, generated by the Higgs–Yukawa coupling $K = y_f v/\sqrt{2}$. The mass M entering the gravitational bulk coupling below is the *total* gravitating mass-energy of a composite body—for ordinary matter, $\sim 99\%$ of which is QCD binding and gluon-field energy in the nucleons, not Higgs-derived quark mass. The two are different quantities

entering at different scales: $K = m_{\text{Higgs}}$ governs the L–R synchronization of a single spinor, while $\Gamma_{\text{grav}} \propto M_{\text{total}}^2$ governs the pairwise gravitational synchronization across the $\sim N^2/2$ constituents of a macroscopic detector. The bridge is the equivalence principle: all forms of energy gravitate, so QCD binding contributes to Γ_{grav} even though it does not contribute to the per-particle Kuramoto coupling K .

The gravitational Kuramoto coupling for a mass M interacting with a bulk at distance Δz is:

$$\Gamma_{\text{grav}} = \frac{GM^2}{\hbar \Delta z} \quad (8)$$

For macroscopic objects ($M \sim 1$ kg, $\Delta z \sim 1$ m), $\Gamma_{\text{grav}} \sim 6 \times 10^{23}$ rad/s—an extremely fast rate, which is precisely why bulk coherence is maintained so effectively for macroscopic masses. The gravitational synchronization coupling grows as M^2 , ensuring that macroscopic objects are locked into collective phase coherence far more strongly than microscopic particles.

4.6 Connection to Penrose–Diósi

This picture has a natural connection to Penrose’s proposal [6] that gravity causes quantum state reduction. Penrose argues that a superposition of two mass configurations with different spacetime geometries is unstable, with collapse timescale $\tau \sim \hbar/E_g$ where E_g is the gravitational self-energy of the superposition.

In the Kuramoto framework, this translates to: the gravitational bulk provides a phase reference, and any quantum superposition that would require the particle to maintain coherence against the gravitational synchronization pressure will decohere on a timescale set by the gravitational coupling strength. The Penrose energy E_g maps onto the Kuramoto coupling rate Γ_{grav} . Both E_g and Γ_{grav} are sourced by total gravitating mass-energy (per the equivalence principle), so the correspondence is between physically commensurable quantities rather than an artifact of notation. The two frameworks predict the same qualitative behavior—heavier objects decohere faster—from different dynamical starting points.

4.7 Geometric Picture: The Coherence Sub-Manifold

The synchronization, interference, and measurement processes described above admit a unified geometric reading. Each Kuramoto phase oscillator carries a phase $\phi \in S^1$, so the joint state of N oscillators lives on the N -torus $T^N = (S^1)^N$. There are not separate manifolds for separate systems; there is one product space whose factors index the oscillators.

Within this product space, every lock condition picks out a lower-dimensional sub-manifold:

- For a single Dirac spinor, the L–R lock $\phi_L - \phi_R = \delta_{CP}$ defines a 1-dimensional curve inside T_{spinor}^2 .
- For the detector bulk, the alignment condition $\phi_1 = \phi_2 = \dots = \phi_N = \Phi_{\text{bulk}}$ defines a 1-dimensional diagonal circle inside T_{bulk}^N .
- For a measured particle, the coupling condition $\phi_A = \Phi_{\text{bulk}}$ defines a sub-manifold inside the joint particle–detector space $T_{\text{particle}}^2 \times T_{\text{bulk}}^N$.

The post-measurement state lies in the *intersection* of these sub-manifolds. The dimensional collapse is dramatic: from $\sim 10^{26}$ free phase coordinates down to effectively one global locked phase.

We refer to this intersection as the **coherence sub-manifold**, because it plays two complementary roles simultaneously:

1. **Dynamical role (attractor).** It is the locus toward which Kuramoto trajectories asymptote. Off the coherence sub-manifold, oscillator phases drift relative to one another; on it, the lock relations are stationary.
2. **Observational role (interference locus).** It is the locus on which coherent phase relations are well-defined. Off the sub-manifold, relative phases $\Delta\phi$ wander ergodically over S^1 and any interference term $\cos(\Delta\phi)$ averages to zero; on it, $\Delta\phi$ is pinned and interference is sharp and observable. The Bell correlation $\cos(2\theta)$ of Section 3 is, geometrically, a function evaluated on the locked sub-manifold of the two-particle phase space—it is well-defined only there.

This dual role yields a useful re-statement of measurement in geometric terms: *the measurement event is not the destruction of interference but the re-routing of the joint trajectory from one coherence sub-manifold to another.* The particle leaves the entanglement sub-manifold (where it shared coherence with its partner) and joins the bulk sub-manifold (where it shares coherence with the detector). What standard accounts call “decoherence” is, in this picture, a change of which sub-manifold the joint state lives on, not a loss of coherent structure as such.

4.8 What This Section Does Not Claim

The re-synchronization picture is interpretive. It does not:

1. Derive decoherence rates quantitatively from Kuramoto parameters (this would require a full quantum treatment of many-body Kuramoto dynamics)
2. The Born rule probability $P = |\alpha|^2$ is addressed in Section 6
3. Provide a Lorentz-covariant formulation of the measurement process

5 Single-World Energy Accounting

5.1 The Energy Bookkeeping Problem

Before measurement, a system $\psi = \alpha|0\rangle + \beta|1\rangle$ carries total energy $E = \hbar\omega$. After measurement produces outcome $|0\rangle$, what happens to the energy associated with the $|\beta|^2$ amplitude?

- Copenhagen: it vanishes—the wavefunction collapses
- Everett: it persists in an inaccessible branch
- Bohmian: the pilot wave carries no energy

Each answer is unsatisfying. We propose a fourth:

5.2 Thermalization of Residual Amplitude

In the Kuramoto measurement process (Section 4), the detector resonantly absorbs the energy quantum $\hbar\omega_0$ corresponding to the realized outcome. The residual amplitude—the portion of the wavefunction that did not synchronize with the detector—has no resonant absorber available.

We propose that this residual couples weakly to the broadband electromagnetic environment and thermalizes as low-frequency radiation:

$$E_{\text{residual}} = (1 - P) \cdot \hbar\omega \tag{9}$$

where $P = |\alpha|^2$ is the Born-rule probability of the realized outcome.

5.3 A Single World

This gives a definite single-world account:

1. ψ is a real physical field carrying energy
2. Measurement is resonant energy extraction (the detector takes $\hbar\omega_0$)
3. The residual thermalizes into the vacuum
4. One classical fact results, with Born-rule probability
5. No branch is created; the residual energy is in this world as thermal radiation

5.4 Honest Caveats

This proposal is the most speculative part of the paper:

1. The thermalization rate is not derived from first principles
2. The mechanism by which residual amplitude “finds” the thermal bath is not specified beyond analogy with spontaneous emission
3. The Born rule is derived from synchronization statistics (Section 6)
4. We do not claim this resolves the measurement problem—it provides a physical picture that is more complete than Copenhagen’s “collapse” but not yet a rigorous derivation

What we explicitly do not claim: We previously explored whether accumulated thermal residuals could account for the observed dark energy density. While a crude order-of-magnitude estimate was suggestive, detailed analysis revealed the calculation to be unreliable, and we withdraw this as a prediction. The thermalization proposal stands independently of any cosmological implications.

6 The Born Rule from Synchronization Statistics

6.1 The Problem

The Born rule— $P = |\psi|^2$ —is the bridge between the mathematical formalism and experimental outcomes. In standard quantum mechanics it is postulated, not derived. Copenhagen asserts it. Everett’s many-worlds claims to derive it from branch counting, but this remains contested. The measurement problem, at its core, is the question of why probability equals the squared modulus of a complex amplitude.

6.2 The Kuramoto Order Parameter

The Kuramoto model provides a natural answer. For N coupled oscillators with phases $\phi_1, \phi_2, \dots, \phi_N$, the degree of collective synchronization is measured by the order parameter:

$$R \cdot e^{i\Psi} = \frac{1}{N} \sum_{j=1}^N e^{i\phi_j} \quad (8)$$

where $R \in [0, 1]$ is the synchronization amplitude and Ψ is the collective phase. The probability that a single oscillator with phase ϕ locks to the collective is proportional to R^2 —the squared modulus of the complex order parameter.

This is not a coincidence. It is a general property of phase-coupled oscillator systems: the coupling efficiency between a single oscillator and a synchronized cluster depends on the squared amplitude of the cluster’s complex coherence.

6.3 Application to Quantum Measurement

Consider a particle in superposition $\psi = \alpha|0\rangle + \beta|1\rangle$ arriving at a detector. The detector is a macroscopic Kuramoto-synchronized bulk with $N \gg 1$ oscillators locked to phase Φ_{bulk} .

The particle carries two components, each a complex amplitude with a definite phase:

- Component $|0\rangle$ with amplitude $\alpha = |\alpha|e^{i\phi_0}$
- Component $|1\rangle$ with amplitude $\beta = |\beta|e^{i\phi_1}$

The Kuramoto synchronization process couples the particle to the detector bulk. The coupling strength between the particle component and the detector depends on the complex overlap—the projection of the particle’s phase oscillator onto the detector’s synchronized state.

For a complex oscillator with amplitude $A \cdot e^{i\phi}$ coupling to a reference oscillator, the synchronization probability is:

$$P_{\text{sync}} \propto |A \cdot e^{i\phi}|^2 = |A|^2 \quad (9)$$

The phase determines *which* detector eigenstate is reached. The amplitude determines *how likely* that synchronization is. Averaging over the random bulk phase Φ_{bulk} (which is uniformly distributed over $[0, 2\pi)$ from the particle’s perspective), all cross terms vanish:

$$\langle e^{i(\phi_0 - \phi_1)} \rangle_{\Phi_{\text{bulk}}} = 0 \quad (10)$$

Only the diagonal terms survive:

$$P(|0\rangle) = |\alpha|^2, \quad P(|1\rangle) = |\beta|^2 \quad (11)$$

This is the Born rule, derived from Kuramoto synchronization statistics rather than postulated.

6.4 Why the Squared Modulus?

The answer is geometric. The wave function is complex because the internal clock is a phase oscillator—it lives on a circle in the complex plane. The probability of synchronization between two oscillators is the squared modulus of their complex overlap because:

1. The coupling is linear in the complex amplitude (Kuramoto sine coupling)
2. Probability is quadratic in the coupling (energy transfer $\propto$ amplitude²)
3. The complex phase averages out over the bulk, leaving only |amplitude|²

This is identical to why the intensity of a classical wave is proportional to the squared amplitude—and it should be, because the wave function *is* a real oscillating field in this framework. The Born rule is the statement that quantum probability equals wave intensity, which is the natural measure for any physical wave coupling to a detector.

6.5 Connection to the Zero-Point Floor

The zero-point phase noise $\sigma_\phi(0) = 1/\sqrt{2}$ rad (Section 9 of the equations reference) sets a minimum uncertainty in the synchronization process. Even at $T = 0$, a particle cannot synchronize with perfect fidelity—the irreducible phase noise means that the probability $|\alpha|^2$ always has a minimum variance. This connects the Born rule to the Heisenberg uncertainty principle: both emerge from the same zero-point phase floor.

6.6 What This Derivation Requires

The derivation rests on three assumptions, all already present in the framework:

1. The wave function is a physical oscillating field (not merely a calculational tool)
2. Measurement is Kuramoto synchronization to a macroscopic bulk
3. The detector’s bulk phase is uniformly distributed relative to the particle’s phase basis

Assumption 3 is the most important. It is equivalent to the statement that the detector has no prior knowledge of which outcome will occur—it is the Kuramoto equivalent of the assumption of measurement independence in Bell’s theorem. The detector synchronizes to whichever component has the largest coupling amplitude, with probability proportional to |amplitude|².

7 Zitterbewegung as a Beat Frequency

Zitterbewegung [3]—the rapid oscillatory motion of a Dirac electron at frequency $2mc^2/\hbar$ —is conventionally attributed to interference between positive and negative frequency components. In the phase-clock picture, it has a more intuitive interpretation.

The temporal clock oscillates at $\omega_t = E/\hbar$. The spatial clock oscillates at $\omega_s = p^2/(m\gamma \cdot \hbar)$. Their beat frequency in the non-relativistic limit:

$$\omega_{\text{Zitter}} \approx \frac{2mc^2}{\hbar} \quad (12)$$

This is exactly the known Zitterbewegung frequency. In the phase-clock interpretation, Zitterbewegung is the **beat note between the temporal and spatial clocks**—observable whenever the two clocks are not perfectly synchronized.

When $K = m$ is large (heavy particle), synchronization is fast and the beat has small amplitude. When K is small (light particle), the clocks are loosely coupled and the beat has larger amplitude. A massless particle ($K = 0$) has permanently unsynchronized clocks at $\theta_{\text{rel}} = 90^\circ$ and no Zitterbewegung—consistent with photons.

8 Antiparticles and CP Violation

8.1 Antiparticles as Time-Reversed Clocks

The negative-energy Dirac solution $v(p, s)e^{+iEt/\hbar}$ has temporal phase running in the opposite sense to a positive-energy particle. In the Kuramoto framework: **antiparticles are particles with time-reversed phase clocks** ($\phi \rightarrow -\phi$).

This is not a reinterpretation—it is already implicit in the Feynman–Stückelberg interpretation [21, 22]. The phase-clock language makes it explicit.

8.2 CP Violation as Synchronization Asymmetry

The Kuramoto equations (2a, 2b) contain the CP-violating phase δ_{CP} . For particles, the equilibrium phase lock is $\Delta\phi = +\delta_{CP}$; for antiparticles, $\Delta\phi = -\delta_{CP}$.

When particle and antiparticle meet for annihilation, their synchronization efficiency depends on the phase mismatch:

$$\sigma_{\text{ann}} \propto |\langle \psi_{\text{particle}} | \psi_{\text{antiparticle}} \rangle|^2 \propto 1 + \varepsilon \cos(\delta_{CP}) \quad (10)$$

The small asymmetry ε accumulates over the early universe to produce the observed baryon asymmetry $\eta \sim 10^{-9}$. This is a qualitative mechanism, not a quantitative prediction—a full calculation requires electroweak baryogenesis beyond the scope of this paper.

9 Extension to Photons via the Riemann–Silberstein Form

Photons are not Dirac particles. They are spin-1 vector bosons described by Maxwell’s equations, with no rest mass to couple chiral sectors. The framework’s central identification $K = m$ therefore does not apply directly: there is no mass term to set a Kuramoto

coupling between photon helicities. We address this by separating two questions that earlier formulations conflated: (i) where do photon Bell correlations come from? and (ii) where does the framework’s measurement mechanism enter for photons?

9.1 The Riemann–Silberstein Form of Maxwell

Maxwell’s free equations can be written in a Weyl-like first-order form using the Riemann–Silberstein vector:

$$\mathbf{F} = \mathbf{E} + i\mathbf{B}, \quad i\partial_t\mathbf{F} = c\nabla \times \mathbf{F}, \quad \nabla \cdot \mathbf{F} = 0 \quad (11)$$

The two helicity eigenstates $\mathbf{F}_+$ and $\mathbf{F}_-$ (right- and left-circular polarizations) are independent—there is no mass term, and the structure mirrors the $m = 0$ chiral limit of the Dirac equation: two decoupled clocks, $\theta_{\text{rel}} = \pi/2$ permanently. A Kuramoto coupling between $\mathbf{F}_+$ and $\mathbf{F}_-$ of a single photon is identically zero in vacuum.

9.2 Where Photon Bell Correlations Come From

The polarization Hilbert space of a photon is two-dimensional, spanned by any orthogonal pair (H/V, $\pm$, L/R). It is mathematically isomorphic to the spin- $\frac{1}{2}$ Hilbert space, with the Poincaré sphere playing the role of the Bloch sphere. For two polarization-entangled photons in the singlet state, the standard quantum correlation is

$$E_\gamma(a, b) = -\cos(2(a - b)) \quad (12)$$

and the CHSH bound saturates at $2\sqrt{2}$. The factor of 2 in the angular dependence reflects the π -periodicity of polarizers (vs. 2π -periodicity of Stern–Gerlach analyzers).

This correlation is a property of the polarization Hilbert space, not of any chiral coupling. It is delivered by standard QED—the same way standard QM with the Dirac spinor delivers the $-\cos(a - b)$ correlation for spin- $\frac{1}{2}$ pairs (Section 2). In both cases, the framework does not derive the Bell correlations; the underlying Hilbert-space structure does. The framework’s contribution lies elsewhere.

9.3 The Measurement Mechanism Transfers to Photons

The framework’s central interpretive claim—measurement is Kuramoto re-synchronization to the detector bulk (Section 4)—does not depend on the $K = m$ identification for its physical content. It requires only that the system being measured carries a phase clock that can be synchronized to the detector. Photons satisfy this: each polarization mode carries a definite phase, and at detection the photon’s polarization clock locks to the polarizer’s selected channel (transmitted vs. rejected).

The Kuramoto coupling for the photon–detector interaction is not zero—it is set by the photon’s interaction with the detector matter (typically dipole coupling $-\mathbf{d} \cdot \mathbf{E}$), not by any vacuum mass term. For a photon of energy $E = \hbar\omega$ coupling to a near-resonant detector channel, the natural Kuramoto rate scales with the photon’s own oscillation frequency:

$$K_\gamma \sim \omega = \frac{E}{\hbar}, \quad \tau_{\text{sync}} \sim \frac{1}{K_\gamma} = \frac{\hbar}{E} \quad (13)$$

A 2 eV optical photon synchronizes to its detector in ~ 0.3 fs; a 10 keV X-ray locks in $\sim 10^{-19}$ s. Higher-energy photons synchronize faster—the Kuramoto rate is set by the photon’s energy via its interaction with the detector, rather than by any rest mass.

This is the photon analog of $K = m$ for fermions: in both cases, the synchronization rate to the detector is set by the energy scale that characterizes the system’s coupling to its environment. For massive fermions at rest that scale is mc^2 (the Compton frequency); for photons it is $\hbar\omega$ (the photon’s own frequency).

9.4 Gravitational Decoherence and the Linewidth-Dependent Bell Test

Two photon time scales must be kept distinct:

- $\tau_{\text{sync}} = \hbar/E$ —the time for Kuramoto re-sync to the detector to complete (set by photon energy);
- $\tau_{\text{coh}} = 1/\Delta\nu$ —the photon’s coherence time during measurement (set by *linewidth*, independent of central energy).

For a given measurement event, the photon is in coherent superposition during the τ_{coh} window. If the two detectors A and B in a Bell test are at different gravitational potentials, their bulk phases $\Phi_{\text{bulk}}(A)$ and $\Phi_{\text{bulk}}(B)$ accumulate a relative offset proportional to $\omega \cdot \Delta\Phi_{\text{grav}}/c^2 \cdot \tau_{\text{coh}}$. When this offset exceeds ~ 1 rad, the singlet correlation degrades.

This is the linewidth-dependent gravitational Bell prediction (`predictions.py` P6b), in self-consistent form:

$$\delta\phi_{\text{grav}} = \omega \cdot \frac{\Delta\Phi_{\text{grav}}}{c^2} \cdot \tau_{\text{coh}} = \frac{\omega \cdot \Delta\Phi_{\text{grav}}}{c^2 \cdot \Delta\nu} \quad (14)$$

$$\text{CHSH}(\Delta\nu) = 2\sqrt{2} \cdot \exp\left(-\frac{\delta\phi_{\text{grav}}^2}{2}\right) \quad (15)$$

Standard QM/QED predicts CHSH is independent of photon linewidth at fixed entanglement fidelity. The framework predicts it decreases with narrowing linewidth in the presence of a gravitational potential difference between the detectors. Tests are in reach using cavity-filtered narrow-linewidth photons at altitude differences of a few km (terrestrial) or with Earth–satellite baselines.

The discriminating measurement is the joint linewidth $\times$ altitude scan with narrow-linewidth photons; photon energy is one knob among three (along with linewidth $\Delta\nu$ and altitude split $\Delta\Phi_{\text{grav}}$), not a discriminator on its own.

9.5 Photons and Penrose Objective Reduction

The gravitational synchronization rate $\Gamma_{\text{grav}} = GM^2/(\hbar\Delta z)$ (Section 4.5) vanishes identically for a massless particle. Equivalently, the Penrose objective-reduction timescale $\tau_{\text{OR}} = \pi\hbar/E_G$ with $E_G = Gm^2/\Delta x$ becomes infinite for $m = 0$. **The gravitational classicalization channel that drives massive systems toward definite outcomes is silent for photons.**

This matches Penrose’s own position. The OR proposal is designed for superpositions of macroscopically distinct *mass* configurations (mirrors, mesoscopic test masses); Penrose has been explicit that single photons fall outside its remit because they have no rest mass to source the gravitational self-energy.

In the framework’s reading, a photon therefore has only two pathways into the classical domain:

1. **Measurement (active).** Kuramoto re-synchronization to a quantized detector channel (§9.3). A single photon hitting a photodetector produces a definite click; this is the only mechanism by which an isolated photon classicalizes.
2. **Coherent-state averaging (many-photon).** A macroscopically populated mode (laser light, classical EM) has a positive Wigner function and obeys Maxwell’s equations classically. The classical limit emerges through statistical superposition of many photon modes, not through any internal collapse mechanism.

A single photon in vacuum, between emission and detection, is **perpetually quantum**—it does not gravitationally classicalize, and it has no way to do so internally. This is consistent with the experimental fact that single-photon Bell tests preserve full coherence over kilometer-scale separations and Earth–satellite baselines, where any internal classicalization mechanism would have produced observable degradation.

One subtlety: photons do follow null geodesics in curved spacetime (gravitational lensing, Shapiro delay). This is unitary classical evolution of the photon’s trajectory, not collapse or decoherence of its quantum state—it bends paths without classicalizing states.

The silence of the gravitational channel for photons is consistent with the linewidth-dependent prediction in §9.4, which attributes the gravitational effect not to the photon in flight but to the bulk-phase mismatch between detectors A and B at different gravitational potentials. The photon is the messenger; the gravitational decoherence is in the detectors.

9.6 What Remains Open

Two questions are not resolved by this section:

1. **A first-principles derivation of K_γ from QED.** We have asserted $K_\gamma \sim \omega$ as the natural scaling, motivated by photon–detector dipole coupling. A rigorous derivation would relate K_γ to the absorption matrix element $\langle e | \mathbf{d} | g \rangle$ for the specific detector channel and compute the actual sync time for realistic photodetectors (photomultipliers, avalanche photodiodes, superconducting nanowire detectors).
2. **The microscopic mechanism for the discrete click.** Section 6 attributes discrete clicks to the threshold structure of detector channels. For photons, this corresponds to the photoelectric or photon-counting threshold of the detection element. The framework provides a mechanism (re-sync to a quantized channel) but does not yet derive the threshold from first principles.

These are tractable extensions, not framework contradictions. The photon-coupling problem—flagged in earlier drafts as the framework’s most serious open question—reduces to a pair of well-defined technical issues once the Bell correlations are correctly attributed to the polarization Hilbert space rather than to any Kuramoto coupling between photon helicities.

10 Qualitative Predictions and Consistency Checks

We distinguish between predictions that follow from the mathematical identification (Section 2) and those that follow from the interpretive claims (Sections 4–6).

10.1 From the Mathematical Identification

P1—Three-term decomposition of Bell correlations. For massive entangled particles, the correlation $E(a, b) = -\cos(a - b)$ decomposes into $E_{LL} + E_{SS} + E_{LS}$, with the relative contributions depending on $\theta_{\text{rel}} = \arcsin(v/c)$. While E_{total} is Lorentz-invariant, the decomposition itself is new and numerically verified. A hypothetical experiment separately addressing large and small spinor components could reveal this structure.

P2—Zitterbewegung as clock beat. The identification of $\omega_{\text{Zitter}} = 2mc^2/\hbar$ as the beat frequency between temporal and spatial clocks is consistent with all known properties of Zitterbewegung. Trapped-ion simulations with tunable effective mass [10] could probe the transition from locked to unlocked clock behavior as effective mass passes through zero.

10.2 From the Interpretive Framework

P3—Decoherence rate scaling with mass. If the Kuramoto synchronization rate is $K = m$, heavier particles should decohere faster: $\tau_{\text{sync}} \sim \hbar/m$. This is qualitatively consistent with the observation that macroscopic objects exhibit essentially instantaneous decoherence while electrons maintain quantum coherence over longer timescales. A precision measurement of decoherence rate as a function of particle mass (electrons, muons, pions, kaons) could test whether $\tau_{\text{decoherence}} \propto 1/m$.

P4—Higgs coupling and synchronization rate. If $K = y_f \cdot \langle \phi \rangle = m$, then decoherence rates for entangled fermion pairs should scale linearly with mass. Comparing entanglement decay times for kaon pairs ($m_K = 494$ MeV) versus B-meson pairs ($m_B = 5,280$ MeV) should show $\tau_B/\tau_K \sim m_K/m_B \approx 0.094$. Existing collider data on neutral kaon and B-meson decoherence could be analyzed.

P5—Gravitationally-weighted secondary-radiation rate. The Stage 2 bulk relaxation (§4.4) emits secondary radiation (virtual photon, real photon, or e^+e^- pair, depending on the energy mismatch) at a rate set by Γ_{bulk} . Because Γ_{bulk} depends on the local mass–energy distribution and is therefore subject to gravitational redshift, the secondary-emission rate during a detection event scales as $1 + \Phi_{\text{grav}}/c^2$ at leading order. Two detectors at different gravitational potentials should register a fractional rate difference of order $\Delta\Phi/c^2$ —roughly 10^{-13} for a 1000 km altitude split. This is below current sensitivity but a clean signature unique to the two-stage framework: neither standard decoherence theory nor Penrose–Diósi predicts gravity-dependent timing of detection-induced secondary emission.

10.3 Honest Assessment

These predictions are qualitative scaling arguments, not quantitative predictions with computed coefficients. The framework in its current form is an interpretation—it provides a physical mechanism consistent with known decoherence behavior but does not yet make

predictions distinguishable from standard decoherence theory. This is the main limitation and the main avenue for future work.

11 Comparison with Existing Frameworks

11.1 Interpretations of Quantum Mechanics

Property	Copen- hagen	Everett MWI	Bohmian	GRW/CSL [23]	Super- determinism	This work
ψ is real?	No	Yes	Yes	Yes	Varies	Yes
Collapse?	Postulated	No	No	Stochastic	Not needed	Re-sync
Branching?	No	Yes (∞)	No	No	No	No
Residual en- ergy?	Discarded	In branches	Pilot wave	Noise	Not ad- dressed	Thermal vacuum
Single world?	By fiat	No	Yes	Yes	Yes	Yes (physics)
Mechanism?	None	None	Pilot wave	Random hits	Initial cond.	Kuramoto sync
Bell?	Accepted	Accepted	Accepted (NL)	Accepted	Denied	Accepted
Born rule	Postulated	Derived?	Derived	Modified	Derived?	Derived (§6)

11.2 Decoherence Models

The closest existing framework is the Penrose–Diósi gravitational decoherence model [6, 7]. Both that model and ours identify gravity as the agent of classicalization. The key differences:

- Penrose–Diósi: collapse timescale $\tau \sim \hbar/E_g$ (gravitational self-energy)
- This work: decoherence timescale set by Kuramoto re-synchronization rate to the gravitational bulk

The predictions are qualitatively similar (heavier $\rightarrow$ faster decoherence) but derive from different dynamics. A quantitative comparison awaits a full derivation of Kuramoto decoherence rates from first principles.

11.3 Zurek’s Einselection

Zurek’s quantum Darwinism [4, 5] identifies “pointer states” as states that survive interaction with the environment. In the Kuramoto picture, pointer states are the synchronized states—the phase-locked configurations that are stable under coupling to the macroscopic bulk. States that cannot synchronize (superpositions of macroscopically distinct mass configurations) are dynamically unstable and decohere. This is consistent with einselection but provides a specific dynamical mechanism (phase locking) for why certain states survive.

11.4 Superdeterminism

Superdeterminism [19, 20] escapes Bell’s theorem by denying measurement independence: the hidden variable λ and the detector settings (a, b) are correlated because they share a common causal past. If $\rho(\lambda|a, b) \neq \rho(\lambda)$, Bell’s derivation does not go through, and local hidden variable models can reproduce quantum correlations.

This comparison is important because the measurement-independence loophole through the gravitational bulk phase Φ_{bulk} —arguing that since both detectors and the particle source share a common gravitational field, their states are correlated—might seem to apply here. We explicitly reject this route in the present paper, as discussed below.

Why this framework is not superdeterministic:

1. **We do not challenge Bell’s theorem.** The quantum correlations $E(a, b) = -\cos(a - b)$ are accepted as arising from the entangled singlet state and the full Dirac spinor structure—standard quantum mechanics. No hidden variable model is proposed to replace them.
2. **Measurement independence is preserved.** The experimenter’s choice of detector angle is not constrained by the framework. The gravitational bulk phase Φ_{bulk} maintains detector coherence (Section 4.4) but does not determine or correlate with measurement settings.
3. **The role of gravity is different.** In superdeterminism, correlations between λ and (a, b) must be fine-tuned to reproduce quantum predictions exactly—a conspiracy that many physicists find unacceptable. In this framework, gravity provides the synchronization bath for the *measurement process* (decoherence), not for the *correlations* (which are quantum mechanical).
4. **No retrocausality or conspiracy.** Superdeterministic models require that the state of the universe at the Big Bang already encodes which detector settings will be chosen in a 2015 Bell test [14]. This framework requires only that macroscopic detectors are phase-coherent objects immersed in a gravitational field—a physically uncontroversial claim.

The distinction can be stated concisely: superdeterminism says the correlations are classical but hidden in the initial conditions. This framework says the correlations are quantum, and the Kuramoto mechanism explains how they are *revealed* (measurement) and *degraded* (decoherence), not how they are *produced*.

12 Discussion

12.1 The Interpretive Status

This paper presents a mathematical identification (Dirac = Kuramoto) and an interpretive framework built on it (measurement = re-synchronization). The mathematical identification is verifiable and, we believe, novel. The interpretive framework is physically motivated but not yet quantitatively predictive beyond standard decoherence.

We are in the category of interpretation papers—alongside many-worlds, Bohmian mechanics, and relational QM [24]. The contribution is not new equations but a new way of reading existing equations that provides physical intuition for the measurement process.

12.2 The Historical Note

The timeline is suggestive:

- 1928: Dirac equation (the two-clock structure)
- 1935: EPR paper [25] (“spooky action at a distance”)
- 1964: Bell’s theorem
- 1975: Kuramoto model

The relativistic spinor structure that produces the full quantum correlation existed 36 years before Bell showed it was needed. The synchronization framework that describes the coupling existed 11 years after Bell. The identification presented here connects these threads.

12.3 Future Directions

Three directions could elevate this from interpretation to testable theory:

1. **Quantitative decoherence rates.** Derive decoherence timescales from Kuramoto parameters in a many-body quantum setting. If these reproduce known rates for specific systems (e.g., superconducting qubits, trapped ions), the framework gains predictive power. If they differ, the framework makes testable predictions.
2. **Born rule—rigorous proof.** Section 6 argues that $P = |\alpha|^2$ follows from Kuramoto synchronization statistics: the squared modulus of the complex coupling amplitude determines synchronization probability, and averaging over the detector bulk phase eliminates cross terms. A rigorous derivation would formalize this in a many-body quantum Kuramoto setting and prove that no other probability measure is consistent with the synchronization dynamics.
3. **Photon coupling.** Resolving the open question of Section 9—what plays the role of K for massless particles—would either extend the framework to all of quantum mechanics or identify its boundary of applicability.

13 Conclusions

We have shown that:

1. **The Dirac mass term has Kuramoto synchronization structure.** In the Weyl basis, the Dirac equation couples ψ_L and ψ_R with coupling constant $K = m$. The Higgs–Yukawa interaction sets $K = y_f \cdot \langle \phi \rangle = m$ exactly.
2. **The NR phase-clock deficiency is traceable.** Retaining only the large (temporal) Dirac component gives $E(a, b) = -\frac{1}{2} \cos(a - b)$. The full four-spinor, with mass coupling $K = m$ ensuring both clocks contribute, restores the exact quantum result through $E = E_{LL} + E_{SS} + E_{LS}$.

3. **Measurement is re-synchronization.** A particle decoheres from its entangled partner by cohering with the detector bulk. The Kuramoto mass asymmetry ($M \gg m$) ensures definite outcomes. Decoherence is not merely loss of coherence—it is redirection of coherence.
4. **The Born rule emerges naturally from synchronization statistics.** The probability $P = |\alpha|^2$ is argued to follow from the squared modulus of the complex coupling amplitude between particle and detector oscillators. The phase averages out; the amplitude squared remains. This is the natural probability measure for any physical wave coupling to a resonant absorber.
5. **Zitterbewegung is the beat frequency** between temporal and spatial phase clocks, at $\omega = 2mc^2/\hbar$.
6. **A single-world energy accounting is possible.** Residual wavefunction amplitude thermalizes into the vacuum rather than branching or vanishing.

The framework does not modify quantum mechanics or challenge Bell’s theorem. It provides a physical mechanism—Kuramoto phase synchronization—for processes (decoherence, measurement, mass generation) that standard quantum mechanics describes but does not mechanistically explain.

In short: quantum mechanics produces the correlations. The Dirac mass term is the Kuramoto coupling that makes particles what they are. And measurement is the moment a small oscillator surrenders its phase to a large one.

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Simulation code and numerical verification available at:

<https://github.com/rayolddog/DiracKuramotoFramework>

The three-term decomposition (Section 3) can be reproduced by running `dirac_extension.py`. The Kuramoto phase dynamics (Section 2) are demonstrated in `higgs_clock.py` and `kuramoto_sync.py`.